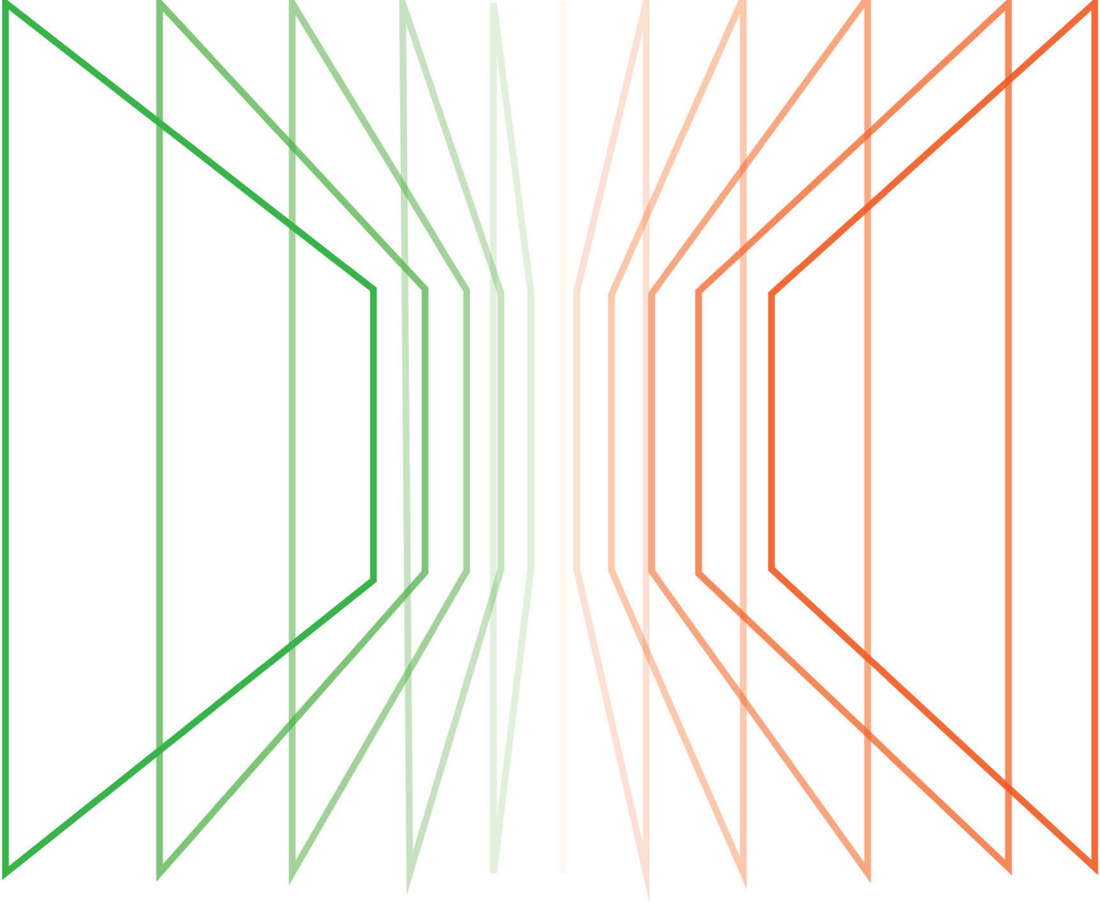
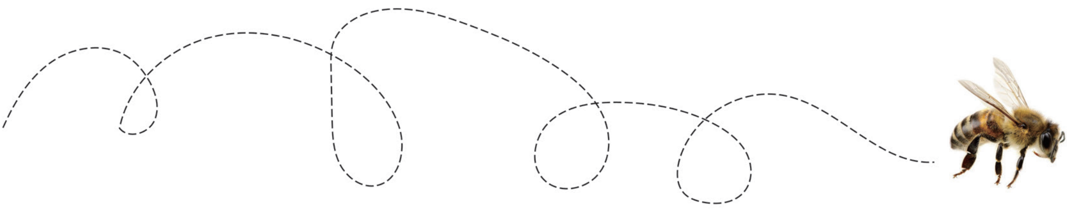


Bee Line



Change Su | MFA Design | 2022

(Title of Thesis)

by ______________

A Project
Presented to
The Graduate Faculty of
California College of the Arts

In Partial Fulfillment
of the Requirements for the Degree
Master of Fine Arts in Design

Approved:

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Chair of CCA MFA in Design Sara Dean

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Thesis Advisor (Type Name)

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Thesis Writing Advisor Mathew Kneebone

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(Additional Advisors, if any)

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date

Content

Digital World

There is a layer upon layer between my world and the real world, with visible and invisible elements intertwined.

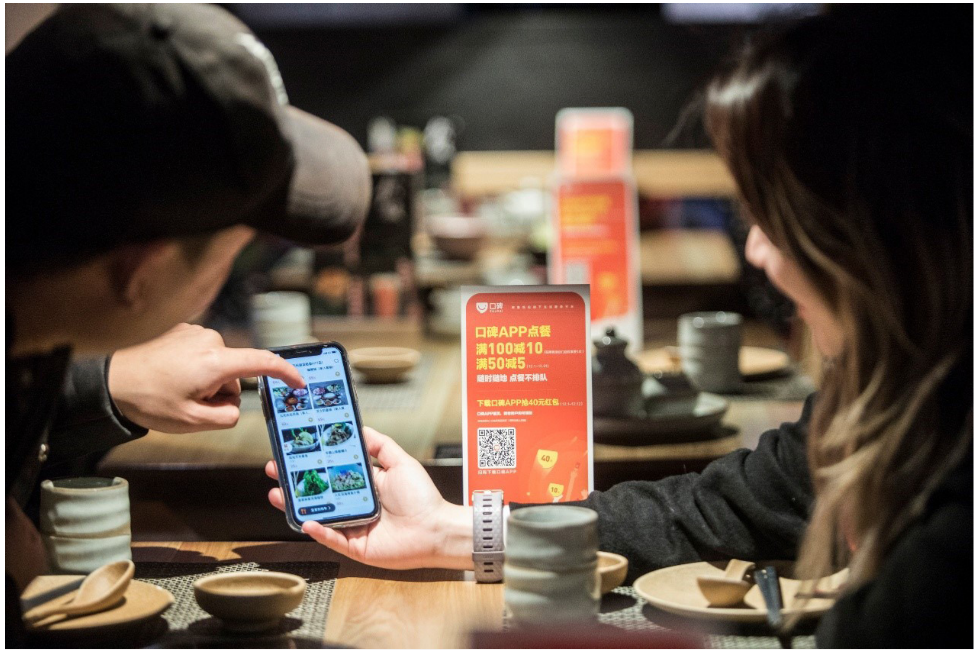
Preface

The world is changing, and many objects are disappearing from our lives in recent years. An object becomes a piece of software in a virtual world. As the Mark Weiser of "Universal Computing" mentioned, the digital world is swallowing the real world. I spent a whole semester looking for my favorite topic, which is around the digital service; smartphones, physical products; daily life objects; uber eats, navigation, and other services. Ordering, navigation, and other topics. I became interested in the relationship between people and the digital world. In one design practice, "Are humans over-indulging in the world behind the screen? What has digitalization taken away from humanity?" I became interested in the relationship between humans and the digital world.

Chapter 01

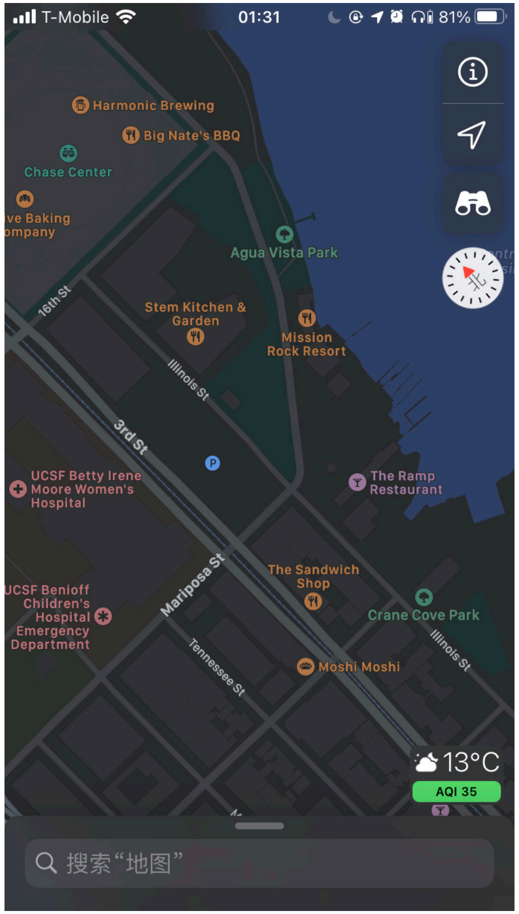
Story and Metaphor——Ordinary Day

In the morning, I wake up by my phone alarm clock. Then, I brush my teeth, wash my face, and prepare to go to work. Before leaving home, I always recheck my bag and make sure my charger and smartphone are still in my bag, cause I need the subway APP to remind me of the arrival time. Meanwhile, before I enter the subway station, and scan the QR code of a shared bicycle to open the lock, which helps me to go easily to the subway station. Then, I will show my Anti-epidemic health code to the testing staff. Scan the QR code to buy a subway ticket. In the end, I end up scanning the company-provided QR code from my phone at the company entrance to enter the office. This story describes my daily life, and I was thinking about how today, with so much digitalization, I only use my phone five-six times at work, but I can barely leave my phone alone all day.

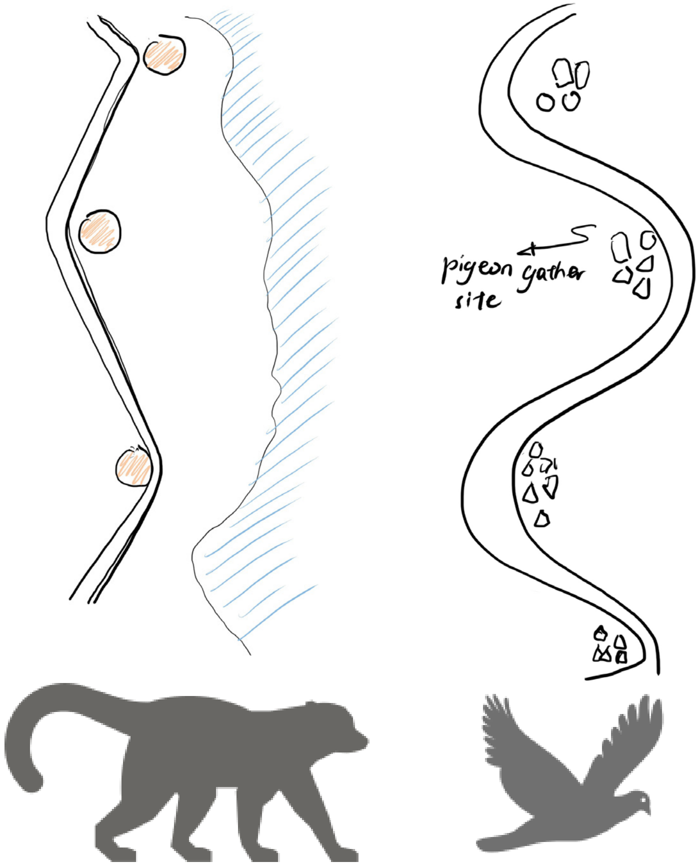


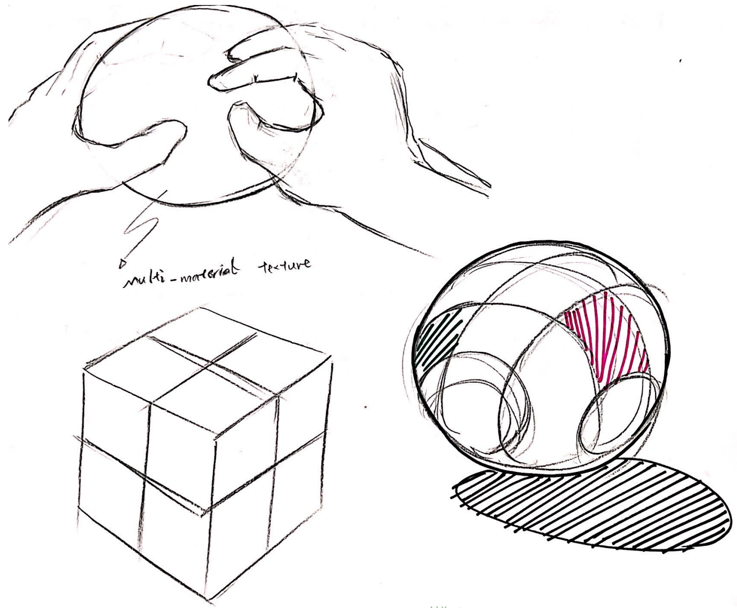
Story and Metaphor——Site-Obstruction

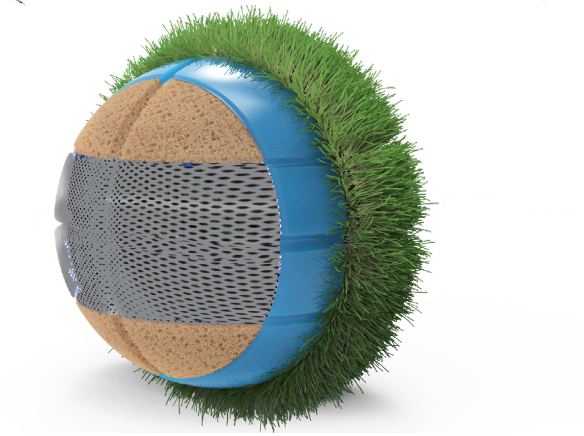
Last semester, I tried to figure out the meaning of screens by researching different digital service software. This project happened in the park. I was using navigation software to check everything about this park. With Google Map, I can see what people are saying about the place, the history of the park, some beautiful scenic locations, and some general directions but as I spent more time in the garden, I discovered that even the seemingly ordinary city parks hide many interesting sights that people don't see every day. For example, there is a broken section of construction fence near the park, and the vast sea breeze blew through these green fabric fences due to the proximity to the sea. Due to different perspectives, when I viewed it up close, I thought it was a crumbling wall, but from a distance, all the holes in the wall connected to show a bird taking off. These exciting observations and interactions with the cityscape came from putting my phone down, getting off the navigation, and getting off the screen. Therefore, I have a conclusion that digital navigation limits the way we explore the real world. We are confined to a small screen, a small square box the size of a palm, an artificial digital world. In this way, we cannot "perceive" the exciting aspects of the real world. Then I started to think about breaking these screens?



Inspired by a raccoon in the park, I created a navigation tool based on raccoon behavior. It allows people to see more at night the way raccoons do. Based on the navigation tool, I propose my thesis core question: What is our navigation? This question leads me to explore navigation methods.





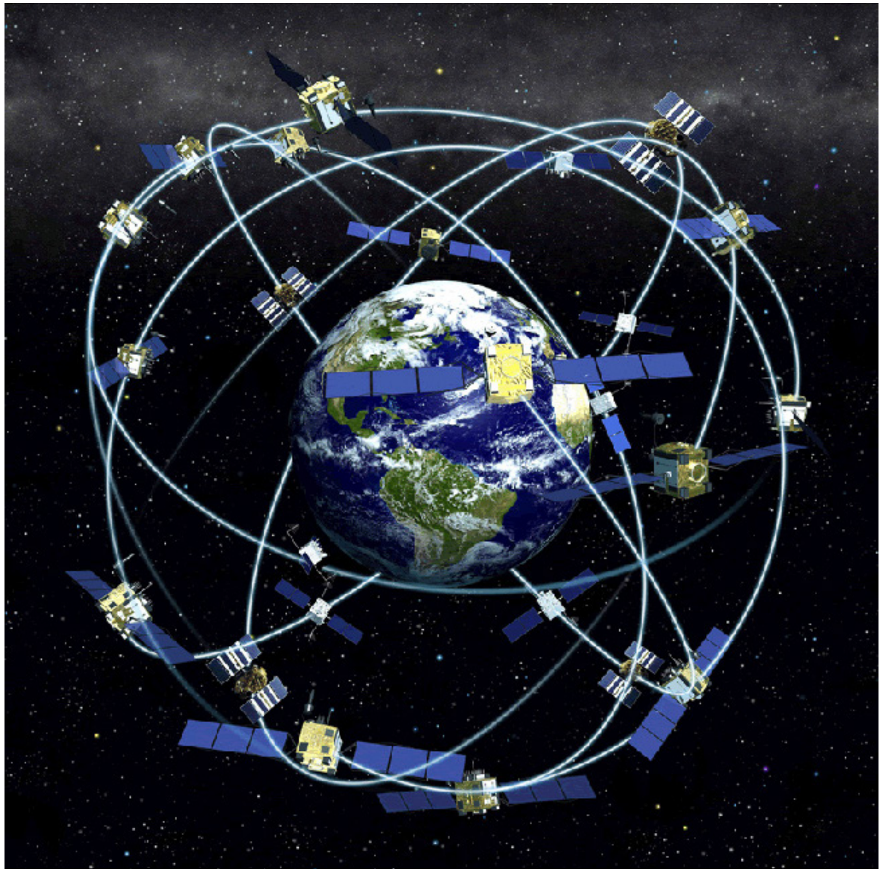


In the following research, this thesis revolves around the question, "What is our navigation? I try to bring animals into the study and practice of design. By drawing on animal perspectives, animal behavior, and animal psychology, I aim to illustrate that there may be alternative ways for humans to use navigation other than GPS-based navigation systems. I think this approach can help people break away from the human-centered design paradigm for a while and think about the value of navigation and its possibilities from a different perspective.

Navigation Software——GPS

Navigation software that provides overly precise directions may not be a good thing. In my research on digital navigation software, I interviewed and observed several people. I wanted to know how they used the navigation software. The word dependence was often mentioned in navigation software. People rely so much on digital navigation software that they can't remember the route after many trips. This is mainly because navigation software often provides obvious ways and tells you approximately how long each course will take. In this case, the dependency situation happens naturally.

The problem with navigation software is not whether it provides accurate routing services but rather the uneasiness that arises when people leave the decision of road selection to the system. There is a crisis of trust in navigation systems. The essence of navigation software is the GPS satellite positioning system. Navigation software is like a wall covering the sky for a person who lives freely on land. We all live under the canopy. GPS satellites watch us like eyes in the sky, sending signals to our cell phones when we need them to tell us where we are. In my experience with navigation software, I found that navigation software does not always give you accurate location. Occasionally it will stop working because of a wrong signal or other reasons. When the navigation software stops working, one can also get confused. Also, following the route provided by the system towards your destination doesn't seem to be what people want. People don't seem to appreciate this service they like in-depend, like when I went to get my vaccinations and was faced with turning on the directions provided by the navigation system or following my roommate home. Yes, I don't seem to trust the navigation system, even though I walked an extra two blocks to get home with my roommate.

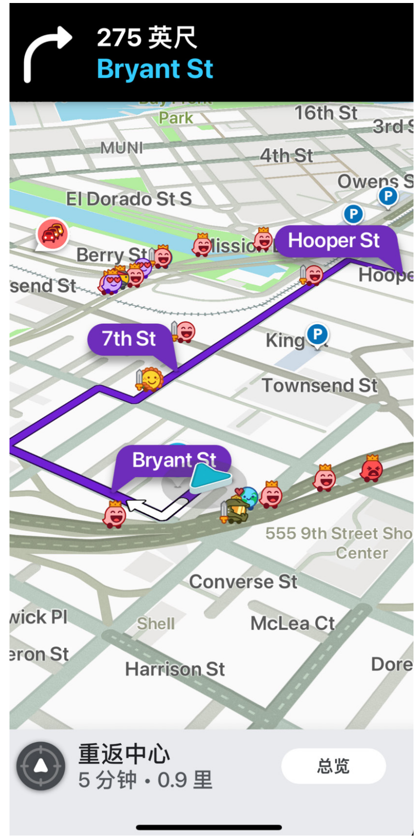


GPS

This is an example. WAZE is one of the most popular navigation software available. It makes road information clearer. It informs drivers about traffic jams; the number of traffic lights; and the location of road repair. But WAZE treats people as cars. It provides traffic data and vehicle data. People trust it on the premise that they obey the rules of the road. However, for pedestrians, the Waze data is meaningless. In this case, the route chosen has a personal characteristic. Navigation software aims to give us a clear route. Navigation software can interrupt a human's process of developing their own wayfinding method preferences, by giving them a bias with a pre-use visual impression.

In this case, I am thinking about how people found their way. Did people find their way? Is navigation software route the best method of finding a course? These questions push me to figure out another way from the history of navigation.

Google Map



WAZE

The lost art of finding our way—History of navigation

The history of navigation is as old as the history of humanity. But the history of GPS is only a short 50 years; humans have existed on the Earth for millions of years. Depending on the cultural background and geographical environment, people develop different ways of finding their way. Inuit living in the polar regions are familiar with using the stars in the sky to find their way. People in the desert often use the sun's position to determine their location. In the primitive jungle, the wet environment of the race will be observed by the root of the moss growth direction to determine the direction. Many clues about orientation exist in the natural environment, and people used to be able to find their way without using GPS navigation. But in recent years, people living in the city seem to have lost this ability.

John Edward Huth observed, the Inuit can draw a precise map of the area where they are located. Hunters could find locations from the prevailing winds due to the preponderance of blizzards and foggy weather. Netsilik hunters keep track of the weather patterns over several days. While simple in concept, the practice of tracking through an ever-changing landscape requires an intuition developed through experience and training. The nature of the tracks tells a story about the passerby.

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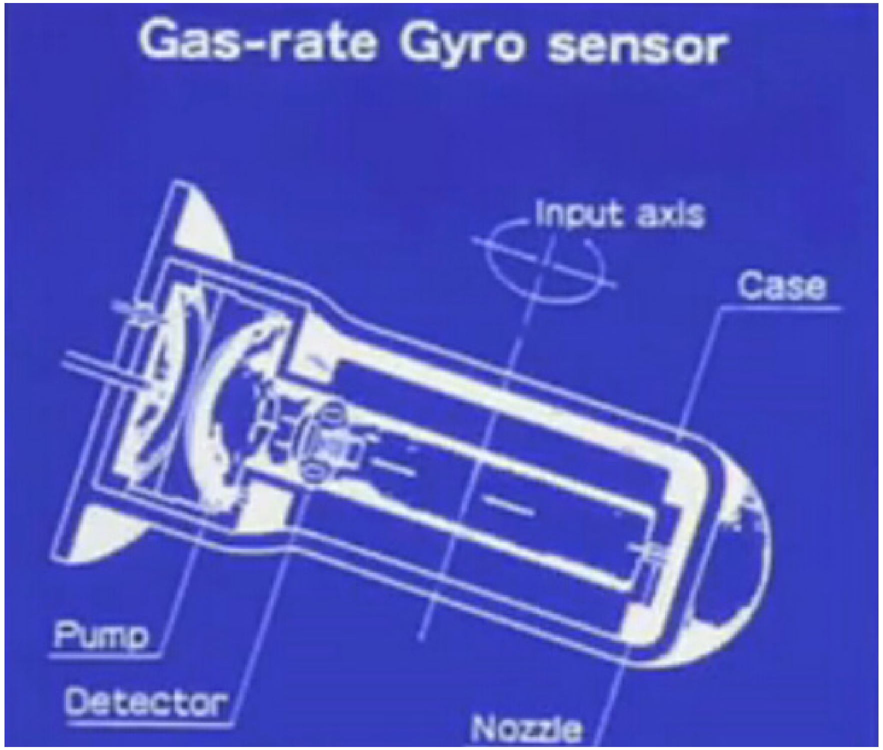
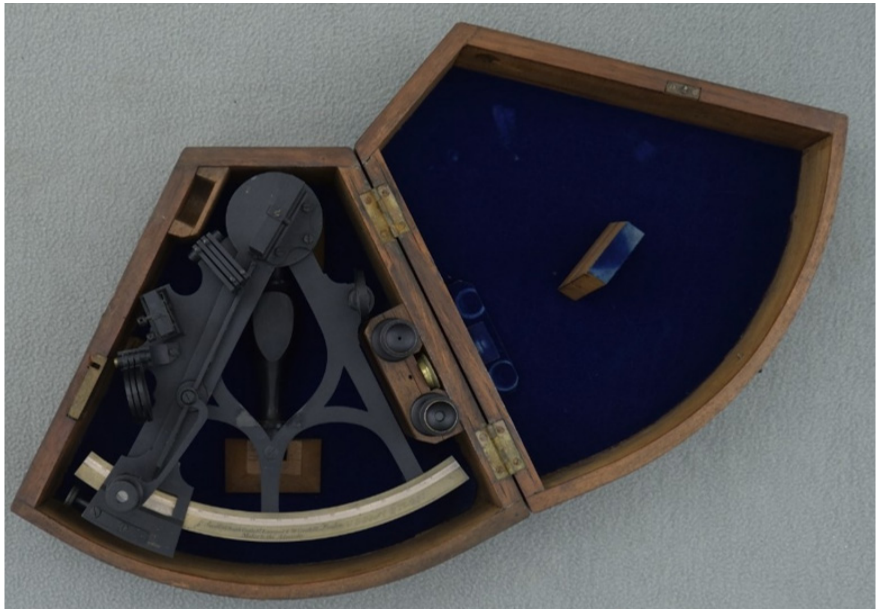


Gyroscope

Sextants for nautical use



Animal Migration and Human Navigation—Finding way as natural roots.

Animal navigation models provide an opportunity to make navigation behavior for human pedestrians more personal. I agree that animal migration is evidence of the natural relationship between the animal and its destination.

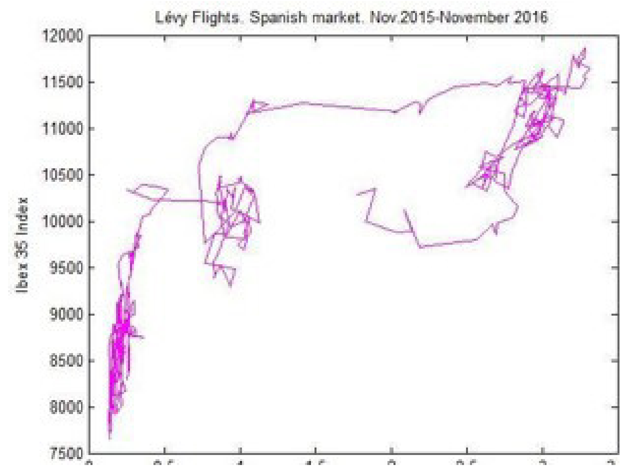
Firstly, animal migration is a common natural phenomenon that has existed for hundreds or thousands of years. Migrating animals are found in all major branches of the animal kingdom. they include taxa as diverse as fish, crustaceans, and amphibians. reptiles, insects, mammals, and slime molds, Interestingly, regardless of the species, they all find their destinations when migrating. In the process of migration, animals find their way according to their own biological characteristics.

Secondly animal Navigation models provide an opportunity to make navigation behavior for human pedestrians more personal. Pedestrians are like animals living in the city. People have many opportunities to observe subtle things in their environment and use them to form their own personal maps. But the bad thing is that the precise routes provided by navigation software obscure these subtle clues and create dependency.

My research builds on the city environment, Thus I focus on the small group of animals living together with humans and figure out how it finds a way by strategy. Animals that depend on navigation are extremely well evolved. They can read much weaker cues than humans. They have multiple alternative strategies. Three animals are my research sample, Bee, Fly, and Cat.

A cat leaves ordos cues and follows these cues to build a personal route. The personal route is patrol.

Such as bees, use the sun compass to find direction, when cloudy the sun, the UV patterns will work. Then, bees closed to their destination, can detect the odors or visual cues to find accuracy. Bee’s navigation method are multiple alternative strategies.



A fly, the fly route pattern is called Levi's flight. Levi's flight is a completely random route pattern, which is why people always fail to catch flies. When animals feel hungry, they use this method to find food.

Conclusion:
In contrast to the three animal navigation modes mentioned above, the navigation software hides some cues that are not visible in the environment, which makes the whole pathfinding process replaced by an overly precise roadmap.

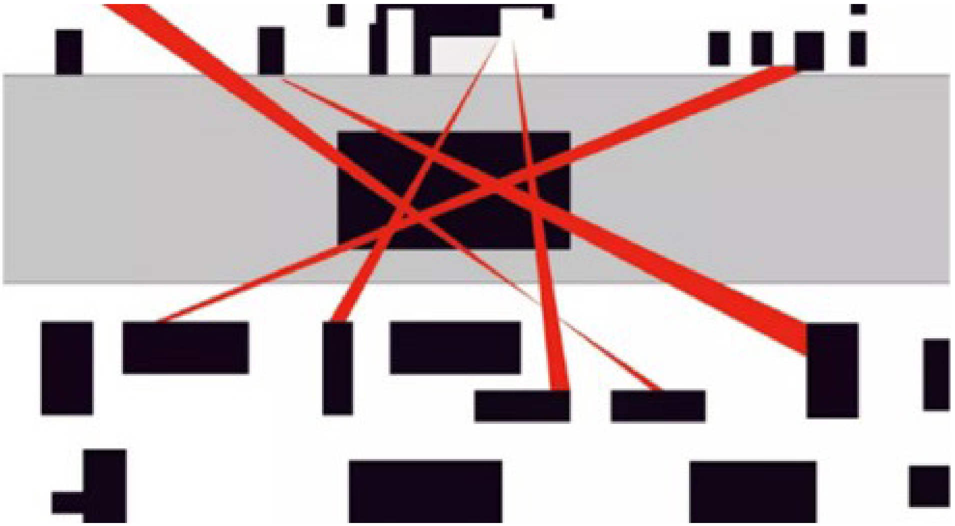
Pedestrians using navigation software=Digital world=Car

Pedestrians without navigation software=Real world=Animal

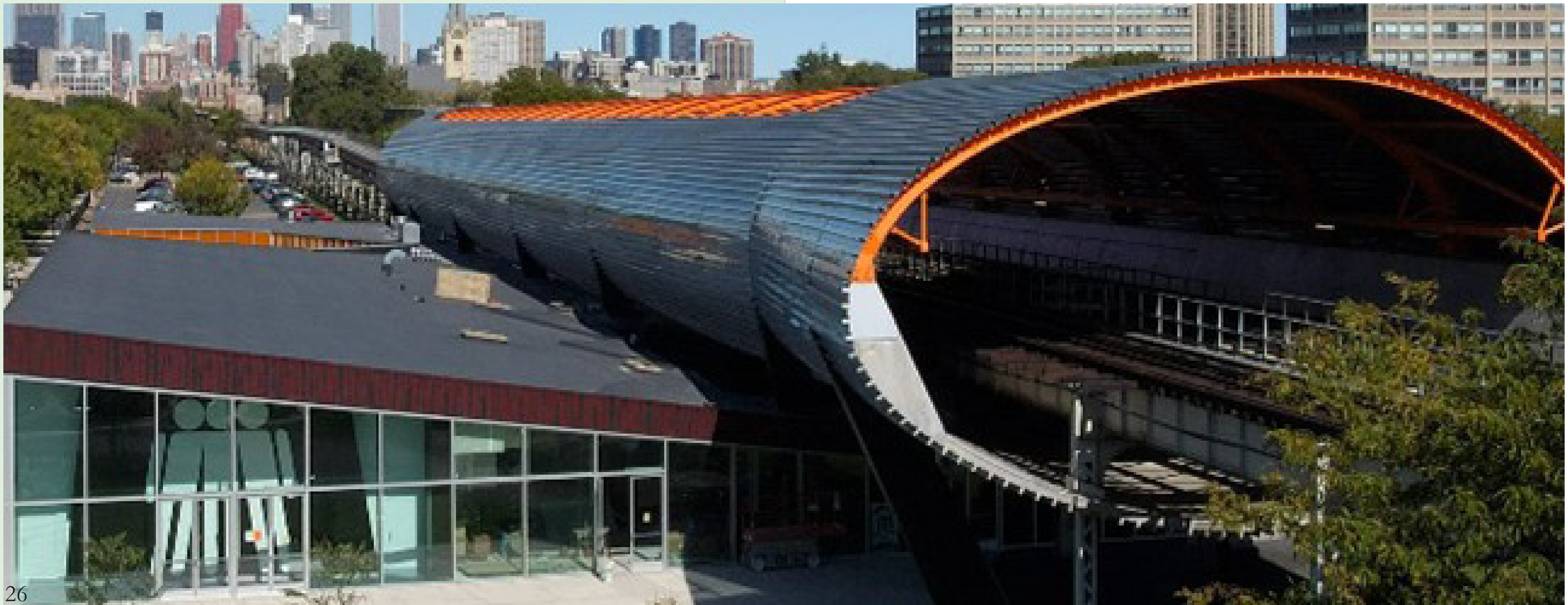
Digital navigation Hides environmental cues.
Animal navigation Uses Environmental Cues.
of the tracks tells a story about the passerby.

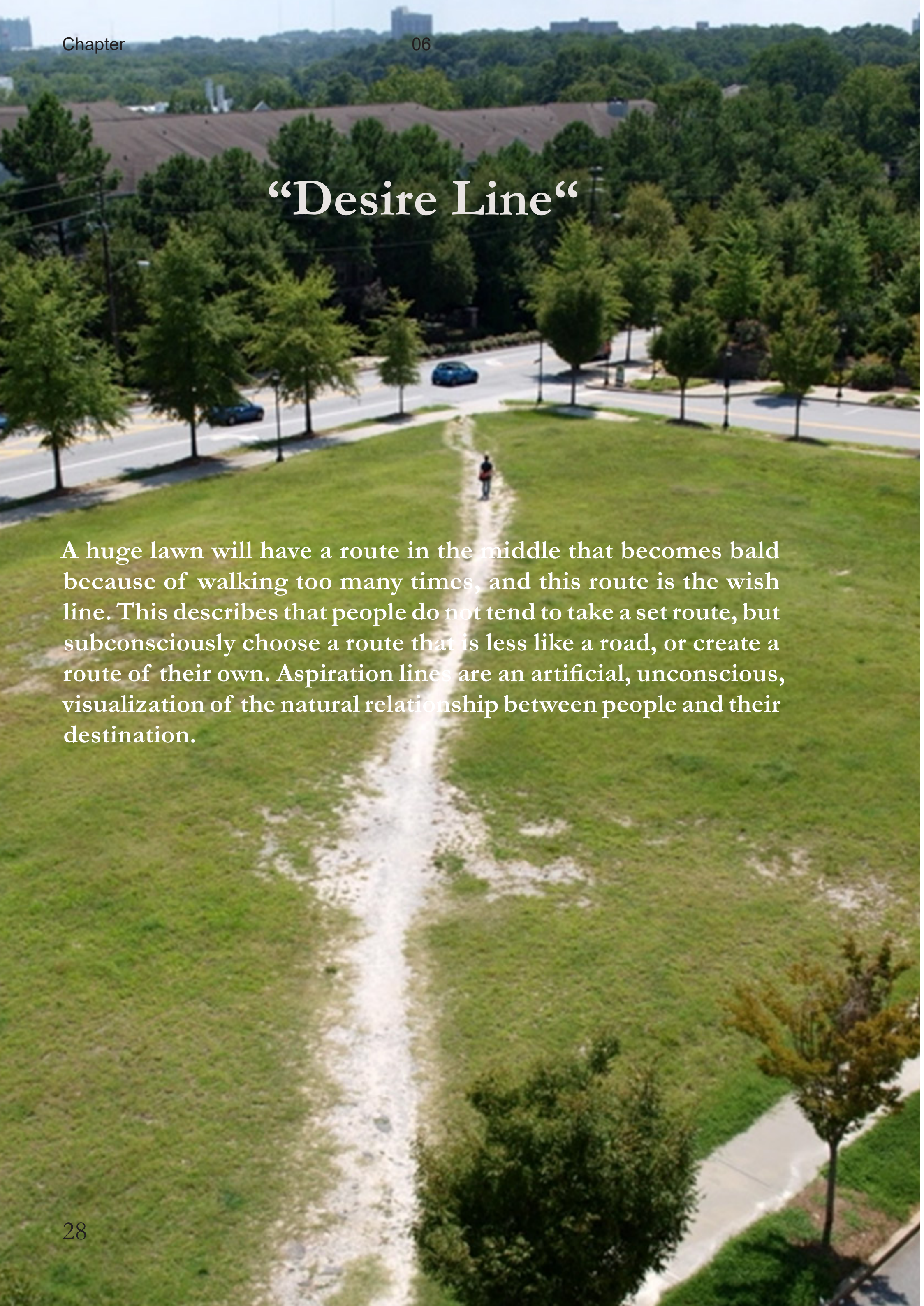
Inspiration & Concept

I got inspiration from the terms "design as participation" and architecture design. IIT McCormick Tribune Campus Center. In 1997 OMA carried out a study to map the "desire lines" of student foot traffic across the campus.



"IIT McCormick Tribune Campus Center"

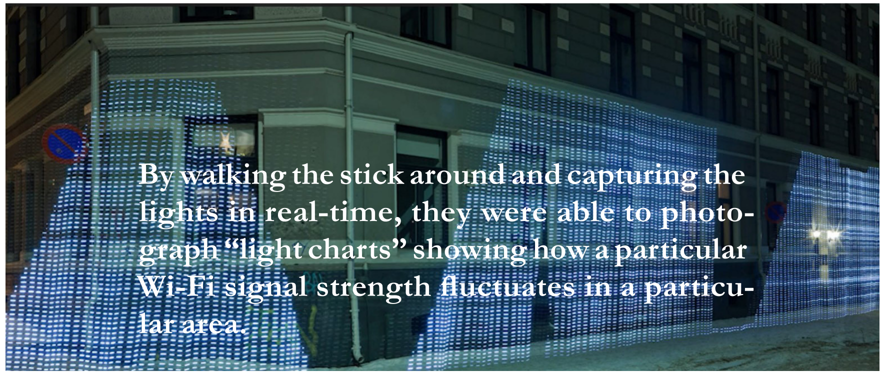




“Desire Line“

A huge lawn will have a route in the middle that becomes bald because of walking too many times, and this route is the wish line. This describes that people do not tend to take a set route, but subconsciously choose a route that is less like a road, or create a route of their own. Aspiration lines are an artificial, unconscious, visualization of the natural relationship between people and their destination.

“Others Work“



By walking the stick around and capturing the lights in real-time, they were able to photograph “light charts” showing how a particular Wi-Fi signal strength fluctuates in a particular area.

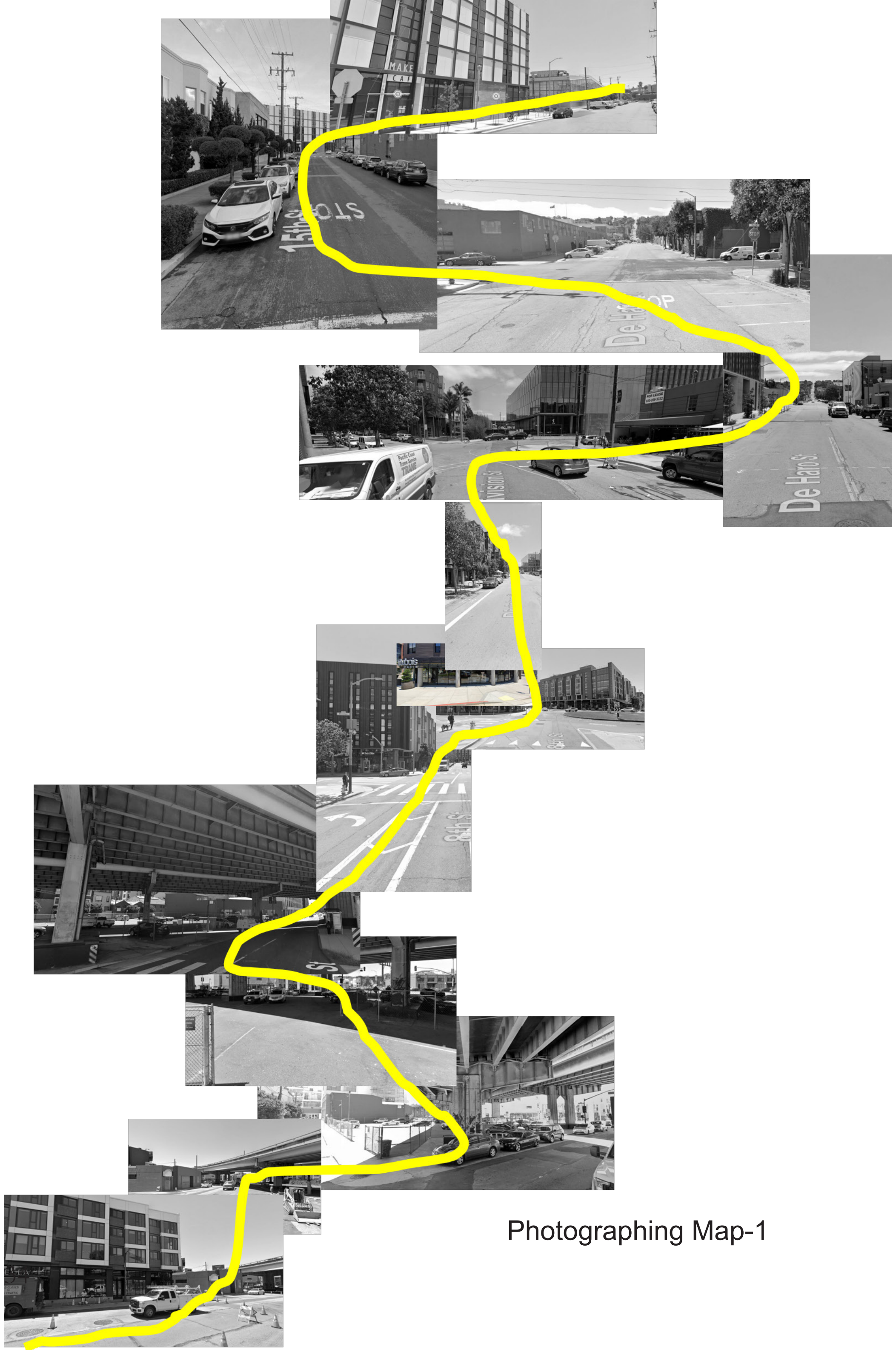
“Building motion heat map video using OpenCV and python”

Recording Map

My concept is from the desire line. However, the desire line is not random creation. I must figure out what elements influence people's behavior in choosing a route. Why do people choose to take the lawn instead of other already obvious paths? Therefore, I create a series of experiments that purpose present the process of creating a desire line and the generation conditions of the desire line. The method of how people extract cues from their surroundings and later make route choices based on the signals, some behavioral details will be shown through the experiment—the investigation site is the route between my home and the CCA campus. The whole journey takes 15 minutes of walking. This is a template for the way. I design the experiment to map a broader range of urban navigation behaviors. All my experiments will be based on this stretch of road as a way to map the more general urban navigation behavior.

Photographing Map.

Photographing while people's attention lingers during walking. The subjects are recombined according to the route. I found that the formation of desire lines may be related to personal memories. The places where people's attention lingers are individual memory points, all of which form a continuous route. These memory points are a form of wish line. Later I reassemble the photos to form a new visual map. The new map disorder the original route to test whether people can find a sense of déjà vu through the confusing images. This way can create a sample, that let other people feel what this road looks like from their home to campus, It is from my home to campus actually.



Photographing Map-1

Photographing Map-2

Architecture Map.

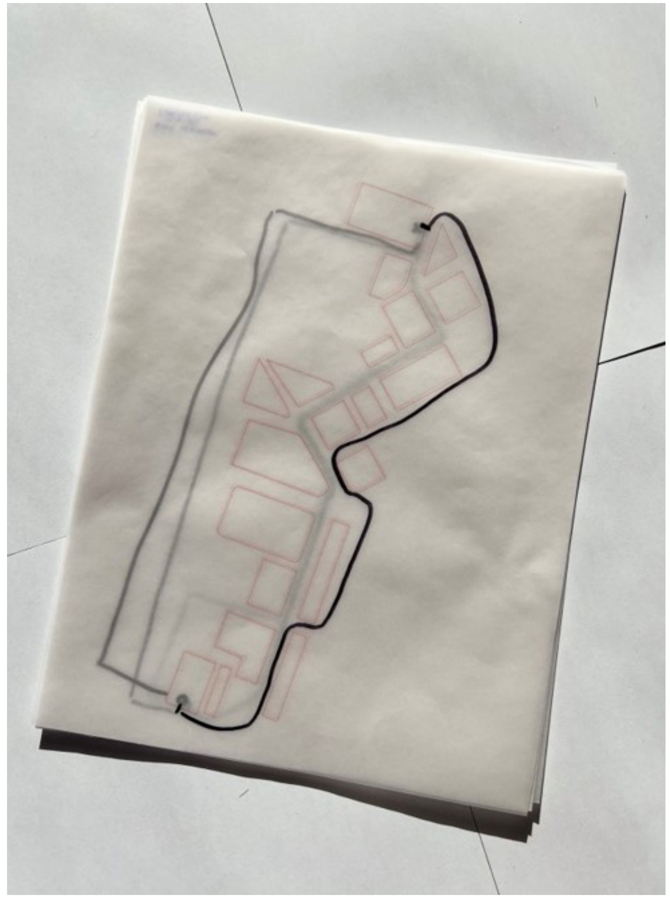
I created a series of maps with only a single clue to show the invisible layers between the digital world and the real world. This is similar to the feeling of a superimposed effect map. This is a map that removes environmental cues and leaves only the buildings around the road. There is evidence of the reality that people orient themselves to landmark buildings. Landmark buildings would also be the most frequent part of people's continuous memory points. Route desire lines dominated by buildings can also reflect people's preferences for route selection. For example, people will choose which one to use as a memory marker based on the height of the building.

Sound Map.

Sound Wave Map. Record the sound made while walking. Use sound as a medium to reflect some detailed changes of people on the road.

Tracing Paper Map.

Using tracing paper, I interviewed the CCA students who live in the same apartment as me to draw the most frequently traveled routes. And ask why. This method was used to find out the reasons for their route choices.



Navigation Design Practice & Psychogeography

These experiments aim to present invisible cues from serious media, which can create new maps. The map seeks to fill the hidden layer between digital navigation software and the real world. These maps are beneficial to build a mental map. Some research shows blind people always use a mental map to recognize their surroundings. In addition, Research on situationist International inspired me to think about the rationality of route selection in navigation. People's choice and understanding of routes are not only influenced by objective reality but also by personal experience, memory, or processing of unique experiences to form new mental maps. The Naked City 1957 is an example of Situationist practice. The Naked City illustrated the central notion in the psychogeography of "centers," sorts of "crossroads" from which several psychogeographical slopes, symbolized by arrows, can be taken.navigation. People's choice and understanding of routes are not only influenced by objective reality but also by personal experience, memory, or processing of unique experiences to form new mental maps. The Naked City 1957 is an example of Situationist practice. The Naked City illustrated the central notion in the psychogeography of "centers," sorts of "crossroads" from which several psychogeographical slopes, symbolized by arrows, can be taken.

Navigation Design Practice & Proprioception

Proprioception often called the "sixth sense" of animals, is a sensory system that detects and controls the position, orientation, and movement of the body. The proprioceptors help the animal to find a specific direction, and it is more like a collection of coordination between the five senses. [c8] A proprioceptive animal can control its body well and quickly perceive the surrounding information and find its direction. Not all human proprioceptors are different from animals. A special case is described in the documentary Proprioception - The Sixth Sense. Athletes who have trained for a long time in teamwork projects tend to be better at functioning as "proprioceptors". They can clearly feel where their body is in space. With your body moving, environmental cues flow around you like water. Compared to animals, humans will have fewer proprioceptive abilities. Therefore, an animal superpower is necessary for humans.

Thus, I tried to put the proprioceptive function of a cat in a human being so that an untrained human can have a "sixth sense" for a short time to choose the right direction. I use Arduino and processing to present my artwork. I used an odor sensor, sound sensor, and light sensor to capture the invisible information in the environment. I use processing to visualize changes in the environment, light, sound, and smell. In addition, I use heart rate sensors to test the change in heart rate when people are affected by environmental factors during pathfinding. assume that the light is colorful particles, which fill the whole space. However, the intensity and angle of the light will change due to the shading of objects in the environment and the passage of time. The particles of light in space will also change. I set the wearer of the device as a magnet. When the person walks down the street wearing a device equipped with a light sensor. As the person moves, the light particles in space are attracted. This is somewhat like people swimming in a pool, and where they swim, the water will follow the force of the person moving. If the water is replaced with a small plastic ball, people will see this walking through. By the same token, the light traces of people walking are thus born. The purpose of this demo is to imitate the superb visual ability of bees.

After testing, I concluded that although the sensors were able to capture information, communicating it through Processing's visual images was not a good option. So I started thinking about how I could better communicate these invisible elements. What would a tool for capturing environmental cues look like? How can devices guide people in the right direction? What would a navigation product with multiple modes look like?

Navigation Design Practice & Artificial Synesthesia

Beeline is an exciting navigation product. It does not provide a clear and unambiguous navigation route, but there is always a direction indicator on the screen pointing you in the right direction to your destination. So you can focus on perceiving your surroundings and immerse yourself in the act of exploration without actually losing track of your target location! Research on artificial association has been going on for many years. In the human world, there are rare phenomena where a small percentage of people can see croaking in blue, or automatically see numbers surrounding their bodies when hearing a continuous string of numbers. Their senses alternate and are interconnected. I took inspiration from this study and applied it to beeline. Using this product as a backdrop, I think about how people pick up environmental cues with the tool? How do people use such navigation tools? Beeline presents its functionality in a visual way.

What is sound vision of Beeline?

ASMR is an electric ear that let people explore tiny sounds from environment.

What is touch vision of Beeline?

Soundwave slippers, pick up the sound in the environment and transform it into vibrations of different frequencies. You can imagine an ear growing on your foot.

What is smell vision of Beeline?

The Flip Flop mimics animal leaves odor cues, I use shoes made of fluorescent materials. Transforming odor traces into visual traces.

What is taste vision of Beeline?

This taste navigation product. It describes the possibility to use the tongue to find the way. It uses an odor sensor to collect air smell, and in scent, the gel could produce basic taste. When people walking in the city, They can use

Each of the 4 product offerings actually refers to an ability of animal navigation.

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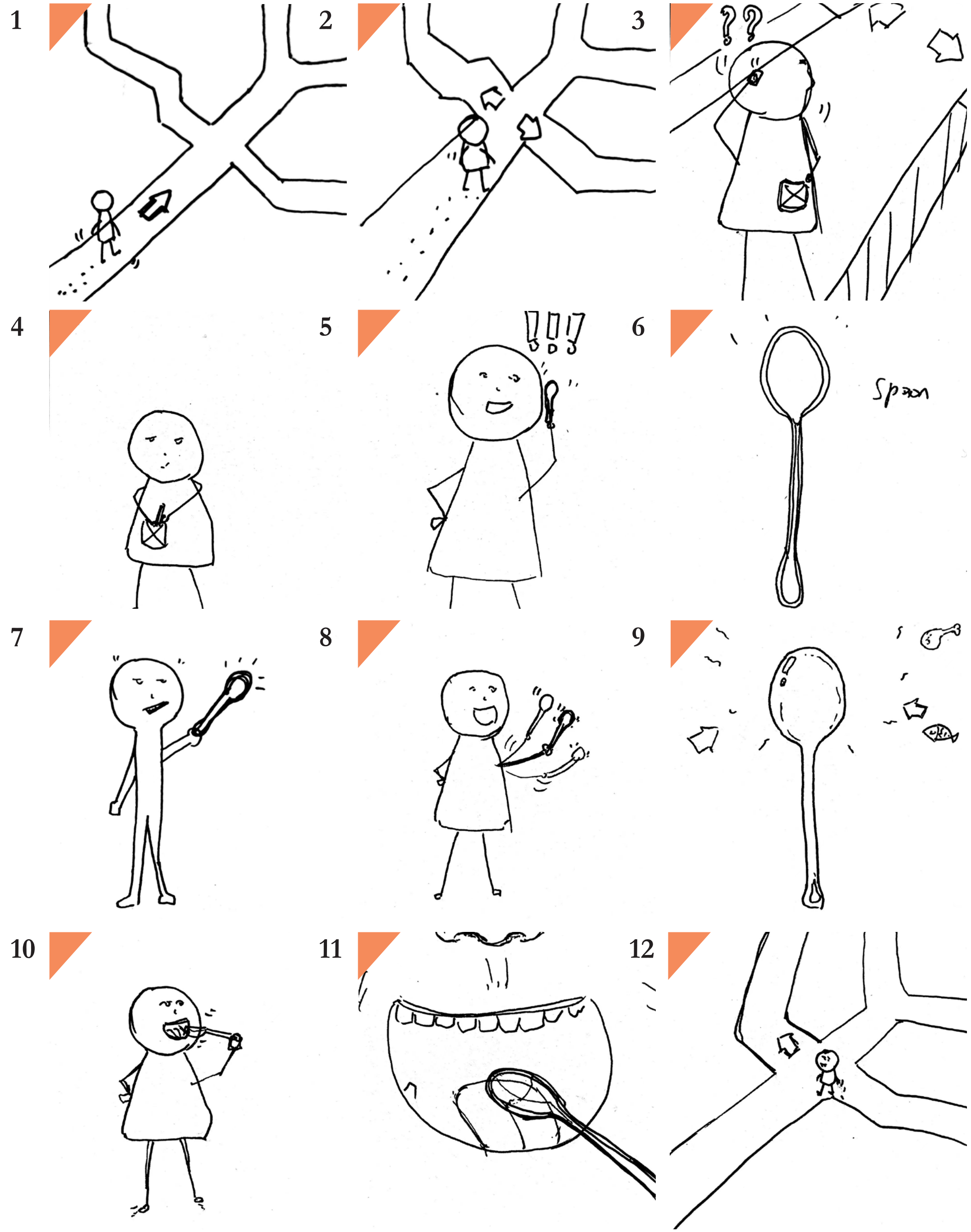

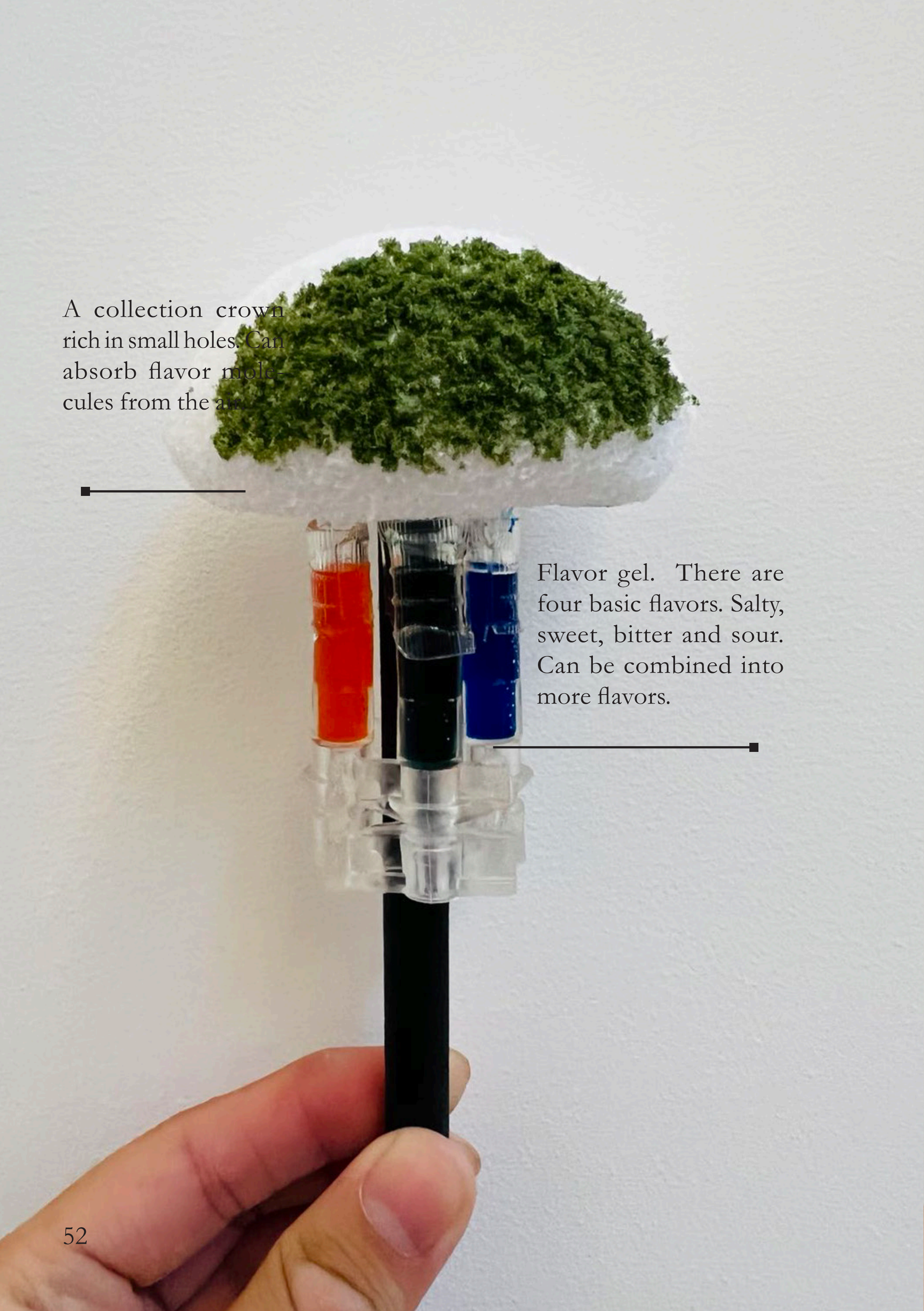

A collection crown rich in small holes. Can absorb flavor mole- cules from the air.

Flavor gel. There are four basic flavors. Salty, sweet, bitter and sour. Can be combined into more flavors.

The flavor formed by the combination of flavor gels will appear in that form above the stomata for easy tasting.

The working condition of the catheter, mixed with the result- ing flavor molecules, will cause the catheter to discolor.

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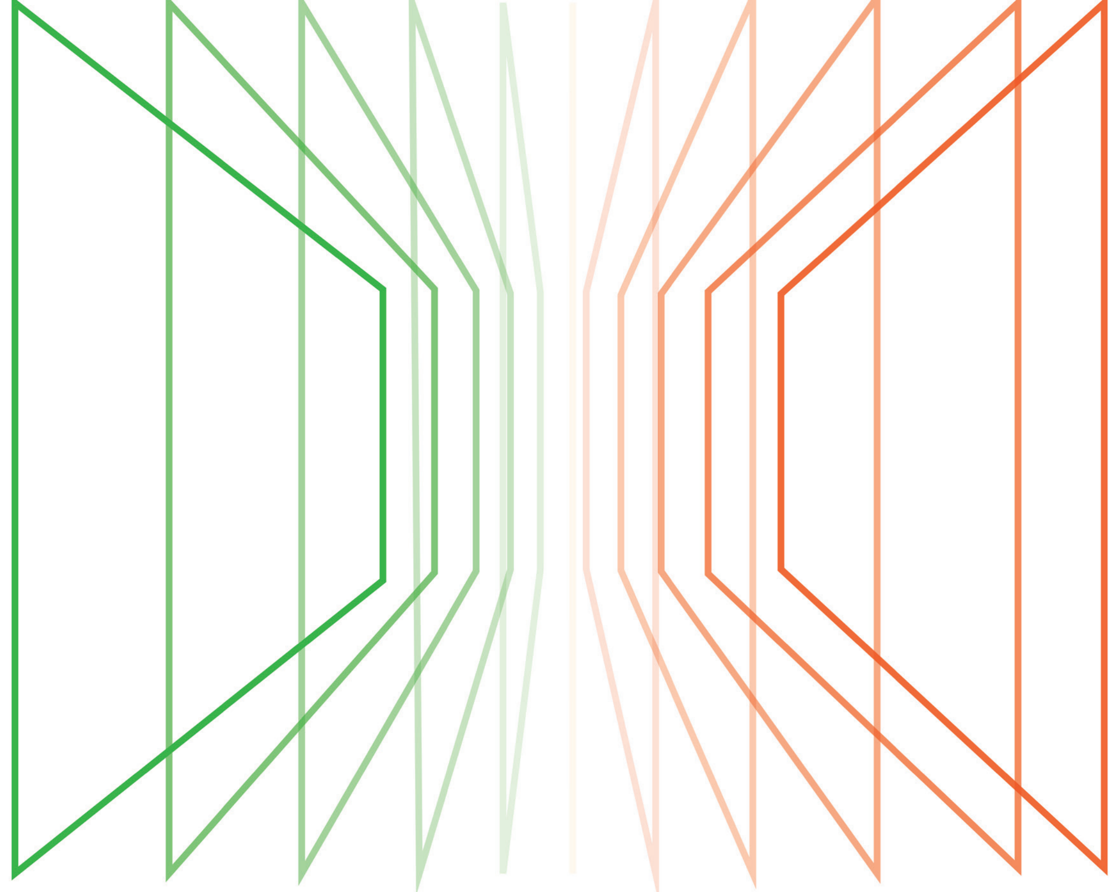
Real World

Where is the distance between me and the digital world?



MFA Design

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